Battery State-of-Charge and Remaining Charge-Time Estimation from Li-ion Drive-Cycle Data

VUDA JAYA SAI GANESH | **Individual project report** | GROUP S2

Abstract

This project estimates the state of charge (SOC) of a Li-ion cell from voltage, current, cell temperature, power and ambient temperature, and the remaining charging time inside charging segments. The frozen dataset has 5,356 rows; a file-grouped split keeps 18 files for training and 5 files for held-out testing so that time-correlated rows never leak between the two. Candidates were compared with grouped cross-validation on the training files and the final model (RandomForest (tuned)) was chosen with a rule fixed before the test set was opened. On the held-out files it reaches MAE 27.73 %, RMSE 37.71 % and R2 -0.417, compared with MAE 18.10 % for Linear regression (raw). Remaining charge time reaches MAE 26515 s on 5 charging file(s) and is indicative only. The complete pipeline is packaged, tested and reproducible from the command line.

1. Introduction and objectives

State of charge (SOC) tells a battery-management system how much usable energy remains, but it cannot be measured directly. This project builds data-driven estimators for (i) SOC in percent and (ii) the remaining time until a charging segment ends, following a strict protocol: a frozen file-grouped split, validation-only model selection, a single scoring of the held-out files and a reproducible software package.

- Estimate SOC from voltage, current, cell temperature, power and ambient temperature.
- Estimate remaining charging time inside constant-current / constant-voltage (CC-CV) charge segments.
- Select the final method from validation evidence, robustness and appropriate simplicity.
- Deliver a frozen, documented and reproducible software package.

2. Data

Source: Li-ion drive-cycle dataset (source / citation to be added via docs/text_overrides.json). The frozen split contains 5,356 rows in 23 files. Features: voltage_v, current_a, cell_temp_c, power_w, ambient_temp_c. Target: soc_percent (SOC, %). The charging target remaining_charge_time_s is the time from each row to the end of its contiguous CHA segment.

Frozen split: GroupShuffleSplit by file_id, test_size=0.2, random_state=42 (Day 12). Training: 18 files / 5,028 rows; held-out: 5 files / 328 rows. Rows from one file are strongly correlated in time, so splitting by file (not by row) is essential to avoid optimistic estimates.

3. Methodology

- Preprocessing: column selection by name only; no scaling; predictions clipped to [0,100].
- Candidates: mean baseline, linear regression, random forest, histogram gradient boosting (an MLP was tried in Week 3 and rejected).
- Selection: GroupKFold CV on train files; within 10% of best CV MAE -> simplest; worst fold <= 2x mean.
- Metrics: MAE, RMSE, R2, plus worst-file MAE and 95th-percentile absolute error; the held-out set is used for the final score only.
- Post-processing: SOC predictions are clipped to 0-100 %.

3.1 Experiment history (Week 3, Day 18)

target	method	features	MAE	RMSE	R2	unit
SOC	Mean baseline	-	23.041	27.509	-0.048	%
SOC	Linear Regression	raw	3.473	4.586	0.971	%
SOC	Linear Regression	engineered	23.193	27.771	-0.068	%
SOC	Random Forest	raw	3.284	10.474	0.848	%
SOC	Random Forest	engineered	22.943	28.917	-0.158	%
SOC	Gradient Boosting	raw	3.301	10.082	0.859	%
SOC	Gradient Boosting	engineered	22.599	27.527	-0.049	%
SOC	MLP (ANN)	engineered	25.932	31.611	-0.384	%
Charge time	MLP (ANN)	charging features	9001.750	18410.150	-0.160	s

Table 1. Methods scored during exploration. Engineered features did not help.

4. Results

4.1 Model selection (grouped cross-validation, training files only)

model	rank	cv_MAE	cv_RMSE	cv_R2	worst_fold_MAE
Mean baseline	0	39.551	41.769	-8.591	44.118
Linear (raw)	1	18.972	29.117	-8.728	29.847
RandomForest (tuned)	2	11.292	20.415	-1.312	26.332
HistGradBoost (raw)	3	10.462	19.363	-1.369	24.125

Table 2. Selection evidence from Day 22.

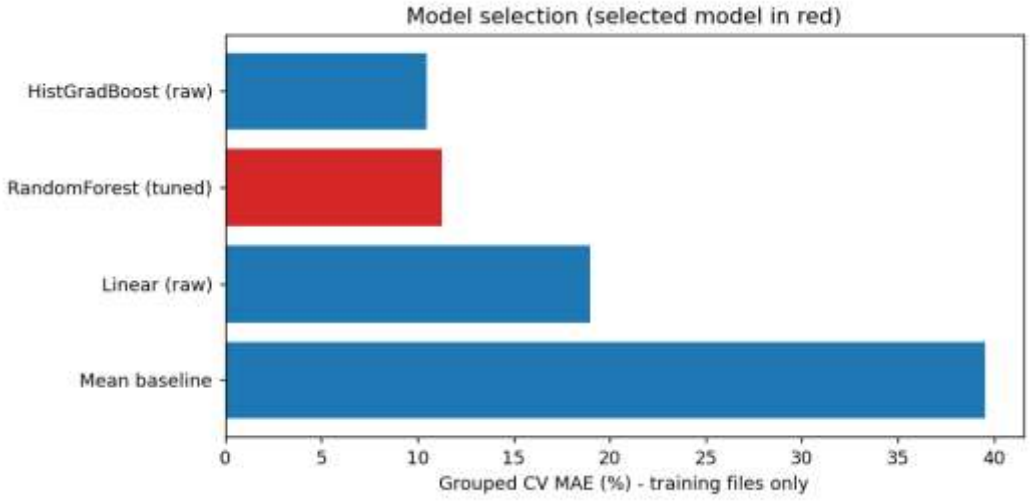


Figure 1. Cross-validated MAE of the candidate models (training files only).

4.2 Final SOC model on held-out files

model	MAE	RMSE	R2
Mean baseline	28.928	32.192	-0.033
Linear regression (raw)	18.104	25.163	0.369
RandomForest (tuned) (final)	27.725	37.708	-0.417

Table 3. Final model versus baselines (held-out files).

The frozen RandomForest (tuned) reaches MAE 27.725 %, RMSE 37.708 % and R2 -0.4171; the worst single file has MAE 35.74 % and the 95th-percentile absolute error is 74.80 %.

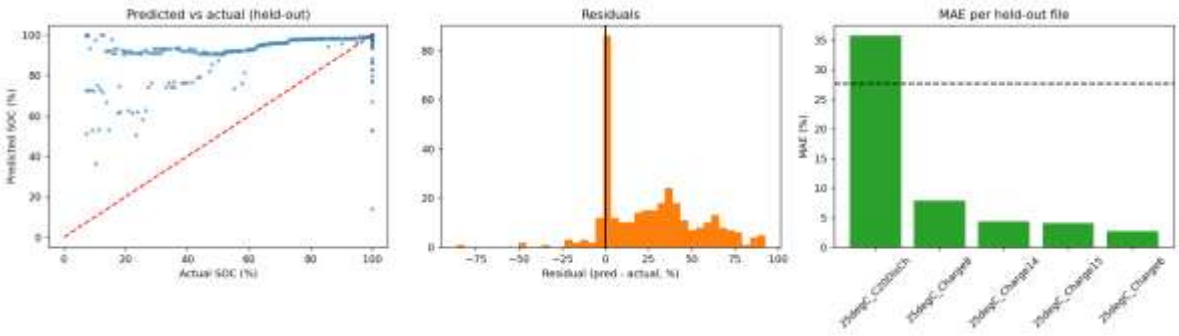


Figure 2. Held-out SOC: predicted vs actual, residuals and per-file MAE.

4.3 Remaining charging time

MAE 26515.5 s, RMSE 34932.5 s, R2 -1.269 on 153 held-out charging rows from 5 file(s). Few files contain charging, so this result is indicative rather than definitive.

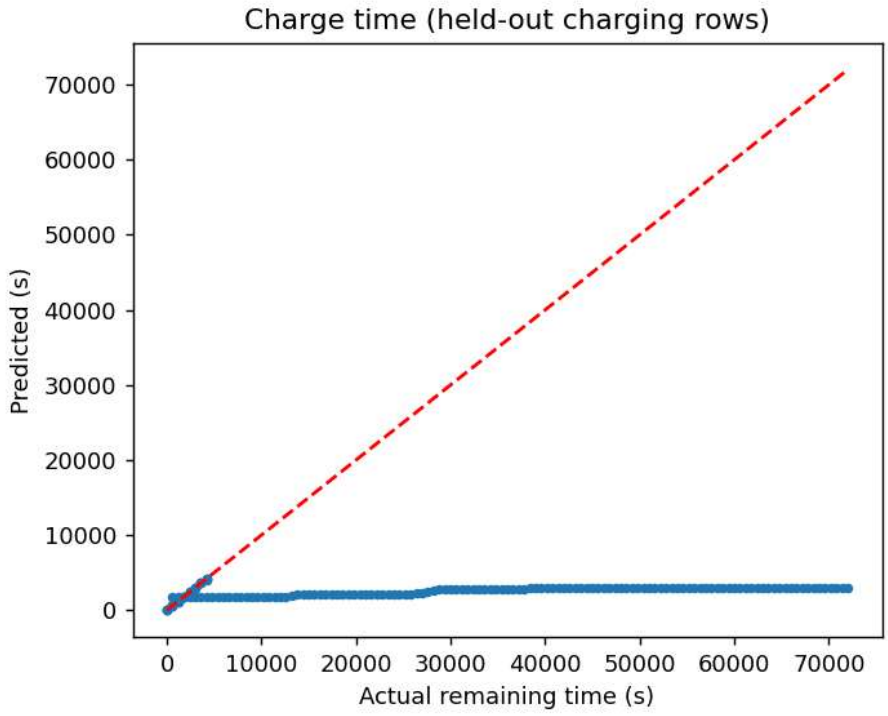


Figure 3. Remaining charge time: predicted vs actual.

5. Error and failure analysis

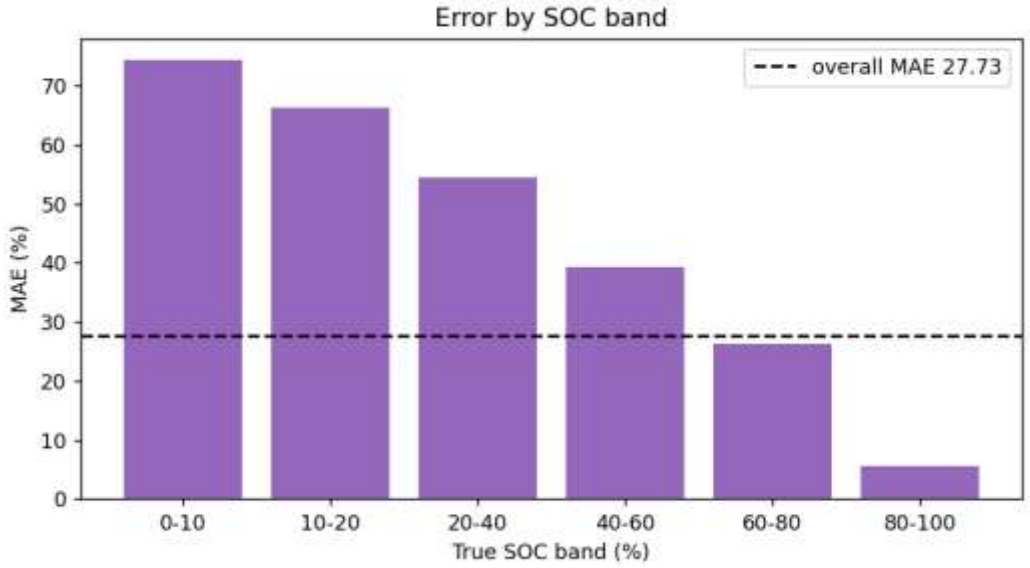


Figure 4. MAE by SOC band (dashed line = overall MAE).

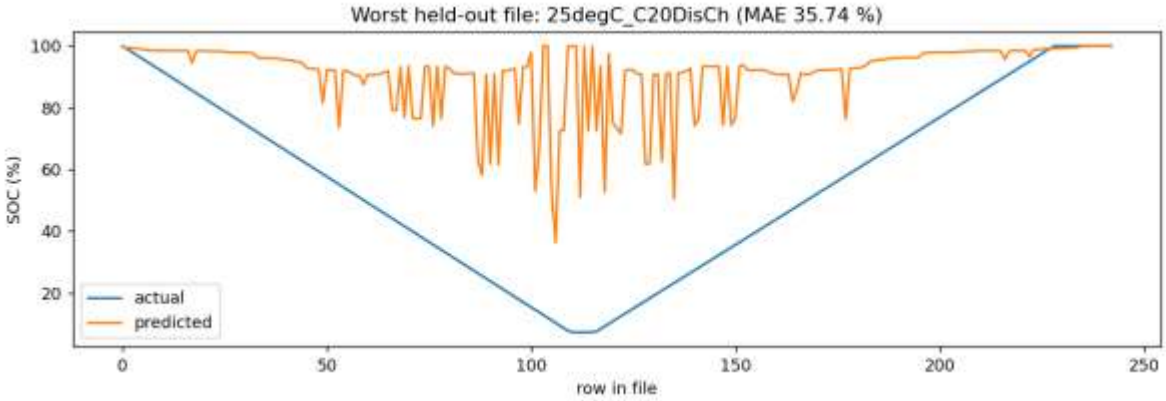


Figure 5. Predicted vs actual SOC for the worst held-out file.

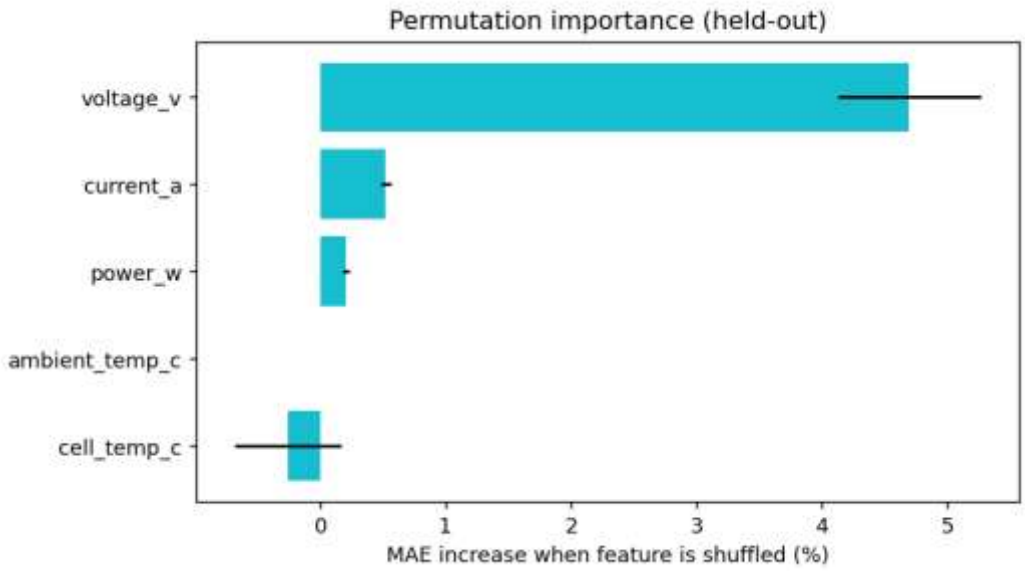


Figure 6. Permutation importance of the input features.

file_id	MAE	max_error	n
25degC_C20DisCh	35.740	92.691	243
25degC_Charge8	7.897	85.954	22
25degC_Charge14	4.332	32.915	21
25degC_Charge15	4.096	47.316	21
25degC_Charge6	2.775	22.446	21

Table 4. Error per held-out file.

5.1 Limitations and likely failure conditions

- Small held-out set - only 5 files / 328 rows, so every metric has wide uncertainty.
- Uneven error across SOC - the highest MAE is in the 0-10 % band (74.33 % vs overall 27.73 %).
- Occasional large errors - RMSE (37.71 %) is 1.4x the MAE; the worst file has MAE 35.74 %.
- Robustness flag from Day 22 - the worst cross-validation fold MAE exceeded 2x the mean fold MAE, so accuracy varies strongly between files.
- Limited operating conditions - ambient temperature in the data: 25 C; other temperatures are untested.
- Static framing - each row is predicted independently (no history), so sensor noise or drift is not filtered.
- Charge-time model is low confidence - evaluated on only 5 file(s).
- Noted on Day 22 - low-SOC knee errors; only 25 degC data; RMSE >> MAE means occasional large mistakes.

6. Reproducibility and software package

Repository: (repository URL to be added via docs/text_overrides.json). The package has the pipeline module, command-line scripts (train / evaluate / predict), the frozen models and configuration, tests, example input/output, figures, tables and this report generator. Retraining from the frozen configuration (same estimator, hyper-parameters and seed) is checked against the stored model.

```
python scripts/predict.py --model-dir models --input
data/sample_input.csv --output outputs/sample_output.csv
```

7. Future scope

- Investigate why the history-based engineered features failed to beat raw features and repair them.
- Add sequence-aware models (windowed features, GRU / LSTM) and compare them under the same grouped protocol.
- Collect data at more ambient temperatures and drive cycles; use leave-one-temperature-out validation.
- Add uncertainty estimates and an out-of-range input detector so unreliable predictions are flagged.
- Collect more charging segments before treating charge time as a production deliverable.
- Extend to other cells / chemistries and to ageing (state of health) effects.

8. Conclusion

A frozen, reproducible pipeline estimates SOC with MAE 27.73 % on unseen files. The main risks are the small number of held-out files, the static (history-free) framing and untested conditions; they are documented with concrete failure conditions and a future-work plan.